

Enterprise Cyber Defense Operations Center (CDOC)

01. Executive Overview

1.1 Project Overview

The Enterprise Cyber Defense Operations Center (CDOC) is a virtualized Security Operations Center environment developed using VMware Workstation. The project simulates an enterprise security monitoring environment using Windows endpoints, Ubuntu, Splunk, Sysmon, Windows Event Logs, Microsoft Defender, Wireshark, and OpenVAS.

The primary objective was to build practical experience in centralized security monitoring, endpoint telemetry collection, SIEM analysis, detection development, network investigation, and vulnerability assessment. The environment was designed to demonstrate how security data from enterprise endpoints can be collected and analyzed from a centralized SOC platform.

1.2 Security Operations Workflow

The implemented environment collects endpoint security telemetry from Windows systems using Windows Event Logs and Sysmon. The collected data is forwarded to Splunk for centralized monitoring, searching, analysis, and detection development. Wireshark is used for network traffic analysis, while OpenVAS is used to identify vulnerabilities within the lab environment.

The implemented workflow is:

Endpoint Telemetry → Log Collection → Splunk Monitoring → Detection & Analysis → Network Investigation → Vulnerability Assessment

1.3 Key Security Capabilities

The completed implementation demonstrates:

- SIEM Monitoring - centralized security event collection and analysis using Splunk.
- Endpoint Monitoring - Windows Event Logs, Sysmon, and Microsoft Defender telemetry.
- Detection Engineering - development and analysis of security detections using Splunk.
- Network Analysis - packet-level investigation using Wireshark.
- Vulnerability Assessment - vulnerability scanning and assessment using OpenVAS.
- Virtual Infrastructure - deployment and networking of the SOC environment using VMware Workstation.

1.4 Project Outcome

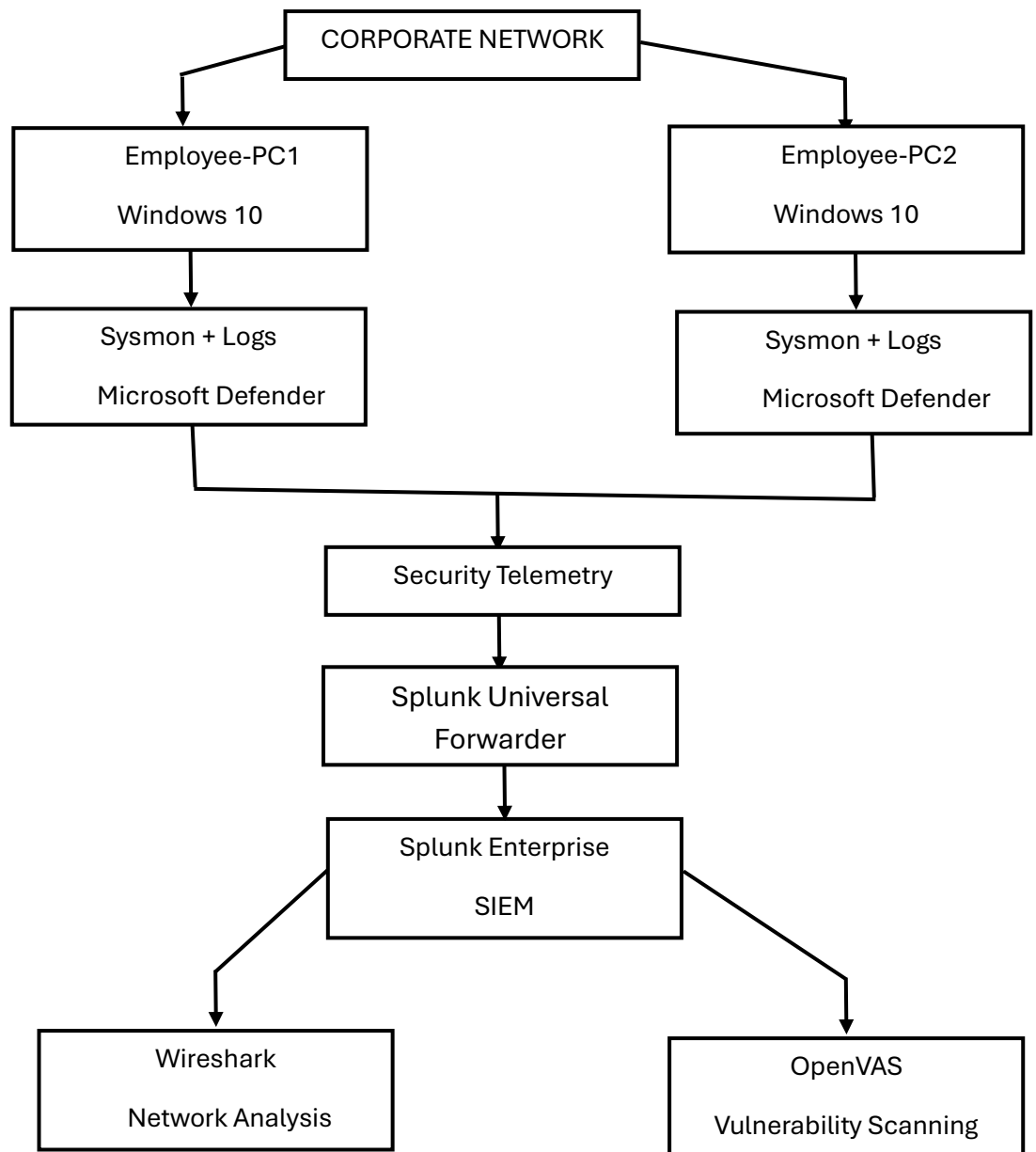
The project demonstrates a practical foundation for SOC operations by combining endpoint monitoring, centralized SIEM analysis, network investigation, and vulnerability assessment within a controlled virtual environment. The implementation provides hands-on exposure to security monitoring and analysis workflows relevant to entry-level SOC Analyst responsibilities.

This project is presented as a controlled laboratory implementation and does not claim to represent a production enterprise SOC.

02. Architecture

The Enterprise Cyber Defense Operations Center (CDOC) is implemented as a virtualized security environment using VMware Workstation. The architecture consists of Windows employee endpoints, a centralized Splunk SIEM environment, endpoint telemetry collection, network analysis, and vulnerability assessment capabilities.

High-Level Architecture



Architecture Workflow

Security telemetry originates from the Windows endpoints through Windows Event Logs, Sysmon, and Microsoft Defender. The Splunk Universal Forwarder collects configured endpoint logs and forwards them to Splunk Enterprise for centralized monitoring, searching, and security analysis.

Splunk provides the primary SIEM layer of the environment, where collected endpoint events can be reviewed and used for security detection and investigation. Wireshark operates as the network analysis component for examining network traffic, while OpenVAS is used separately for vulnerability assessment of the lab environment.

Core Data Flow

Windows Endpoint → Sysmon/Windows Logs → Splunk Universal Forwarder → Splunk Enterprise → Security Monitoring & Analysis

Wireshark provides network-level visibility, while OpenVAS provides vulnerability assessment capabilities. These components represent different areas of the SOC workflow without claiming integrations that were not implemented.

The architecture therefore provides a controlled environment for practicing endpoint monitoring, centralized SIEM analysis, network investigation, and vulnerability assessment using multiple security technologies.

03. Technology & Infrastructure

The CDOC is implemented as a virtualized security operations environment using VMware Workstation. The infrastructure provides a controlled environment for endpoint monitoring, centralized SIEM analysis, network investigation, and vulnerability assessment.

3.1 Virtual Infrastructure

VMware Workstation provides the virtualization layer for the laboratory. It hosts the Windows endpoints and Ubuntu environment, allowing the security components to operate within a controlled virtual network.

3.2 Enterprise Endpoints

Employee-PC1 and Employee-PC2 represent Windows 10 employee workstations within the simulated enterprise environment. These endpoints generate the security telemetry used for monitoring and analysis. Windows Event Logs, Sysmon, and Microsoft Defender provide visibility into endpoint activity and security events.

3.3 SIEM and Log Collection

Splunk Enterprise acts as the central Security Information and Event Management platform. It receives endpoint telemetry and provides centralized log searching, event analysis, detection development, and security monitoring.

The Splunk Universal Forwarder is used to collect and forward configured Windows and Sysmon logs from the monitored endpoints to Splunk Enterprise.

3.4 Endpoint Telemetry

Sysmon provides detailed endpoint event information that supports security monitoring and detection development. Windows Event Logs provide native operating system security events, while Microsoft Defender provides endpoint security protection and related telemetry. These sources provide the security data analyzed within Splunk.

3.5 Network and Vulnerability Analysis

Wireshark is used for packet-level network traffic analysis and investigation. OpenVAS is used to perform vulnerability scanning against the lab environment and identify security weaknesses requiring attention.

3.6 Technology Stack

SOC Function	Technology
Virtualization	VMware Workstation
Endpoint Monitoring	Windows, Sysmon, Microsoft Defender
SIEM	Splunk Enterprise
Log Collection	Splunk Universal Forwarder
Network Analysis	Wireshark
Vulnerability Assessment	OpenVAS

Together, these technologies form the implemented foundation of the CDOC for centralized security monitoring, endpoint analysis, network investigation, and vulnerability assessment.

04. Endpoint + SIEM

The CDOC uses a centralized endpoint monitoring and SIEM architecture to collect and analyze security activity from the simulated Windows enterprise environment. Employee-PC1 and Employee-PC2 act as monitored Windows 10 endpoints and provide the primary security telemetry used within the lab.

4.1 Endpoint Telemetry

The endpoint monitoring layer uses Windows Event Logs, Sysmon, and Microsoft Defender. Windows Event Logs provide operating system and security events, while Sysmon provides detailed endpoint telemetry for security

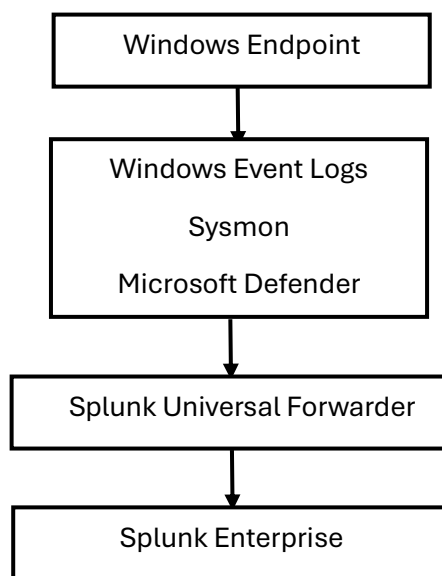
monitoring and detection analysis. Microsoft Defender provides endpoint protection and additional security telemetry.

The collected telemetry provides visibility into activities including:

- Process creation and execution
- PowerShell activity
- Authentication events
- DNS activity
- Network connections
- Account-related events
- Endpoint security events

4.2 Log Collection and Forwarding

The Splunk Universal Forwarder collects configured endpoint logs and forwards them to Splunk Enterprise for centralized analysis.



4.3 Splunk SIEM

Splunk Enterprise functions as the central SIEM platform. The collected telemetry can be searched and analyzed from a centralized interface, supporting security monitoring and detection development.

The implemented monitoring and detection work covers activities including suspicious PowerShell, encoded PowerShell, local user creation, administrator-group changes, Defender-related activity, process creation, DNS activity, and network connections.

4.4 Operational Value

Centralizing endpoint telemetry in Splunk allows related events from different sources to be reviewed together during security analysis. This provides greater visibility than examining individual endpoint logs separately.

The endpoint-to-SIEM architecture forms the core monitoring layer of the CDOC and demonstrates practical experience with Windows security telemetry, Sysmon, log forwarding, Splunk SIEM, and security event analysis.

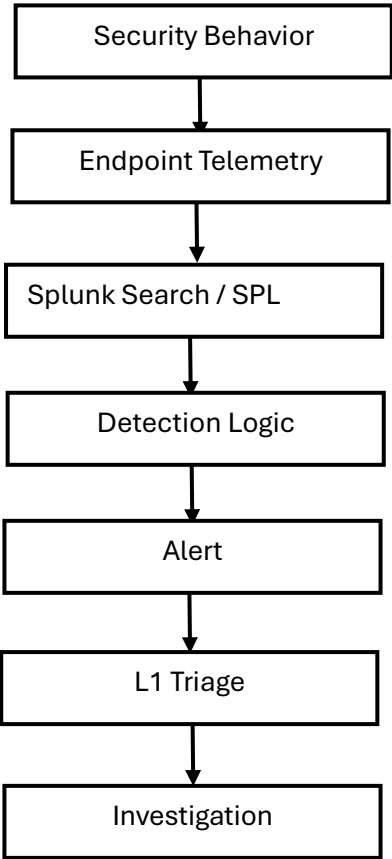
05. Detection Engineering

Detection engineering within the CDOC focuses on converting endpoint telemetry into actionable security detections. The objective is to identify security-relevant behaviors, generate useful alerts, and provide sufficient

context for SOC investigation. The project implements detection use cases in Splunk using telemetry primarily collected from Windows Event Logs and Sysmon.

5.1 Detection Workflow

The detection process follows:



Detection logic is based on observable security behaviors, allowing collected telemetry to support both alert-based detection and manual investigation.

5.2 Implemented Detection Use Cases

Detection	Primary Telemetry	Purpose
Suspicious PowerShell	Sysmon	Identify potentially abnormal PowerShell execution
Encoded PowerShell	Sysmon	Detect potentially obfuscated PowerShell commands
Local User Creation	Windows Security Logs	Identify unexpected account creation
Administrator Group Modification	Windows Security Logs	Detect privileged-group changes
Defender Tampering	Windows/Defender Telemetry	Identify activity that may weaken endpoint protection
Suspicious Process Activity	Sysmon	Identify unusual process execution
DNS Activity	Sysmon	Investigate potentially suspicious domain resolution
Network Connections	Sysmon	Identify potentially abnormal outbound connections

These detections cover execution, account and privilege management, defense evasion, and network activity.

5.3 Detection Development

For each detection, the relevant telemetry source is identified and an SPL query is developed to isolate the behavior of interest. Results are reviewed to determine whether the detection produces meaningful events and whether legitimate administrative activity could generate false positives.

Detections are evaluated based on data source, detection logic, severity, false positives, investigation context, and recommended next investigative action.

5.4 Analyst Investigation

When a detection generates an alert, the analyst validates the activity rather than automatically classifying it as malicious. The affected endpoint, user, process, command-line information, timestamps, and related events are examined. Additional telemetry can be correlated in Splunk to establish an activity timeline and determine whether the behavior is legitimate, suspicious, or requires further investigation.

This demonstrates the transition from basic log collection to detection engineering and analyst-driven investigation within the CDOC.

06. Monitoring + Threat Hunting

The CDOC combines security monitoring with proactive threat hunting to identify suspicious activity across the simulated enterprise environment. Splunk Enterprise serves as the primary monitoring and investigation platform, using endpoint telemetry collected from Windows Event Logs, Sysmon, and Microsoft Defender.

6.1 Security Monitoring

SOC monitoring focuses on identifying abnormal or security-relevant activity across Windows endpoints. The collected telemetry allows analysts to monitor:

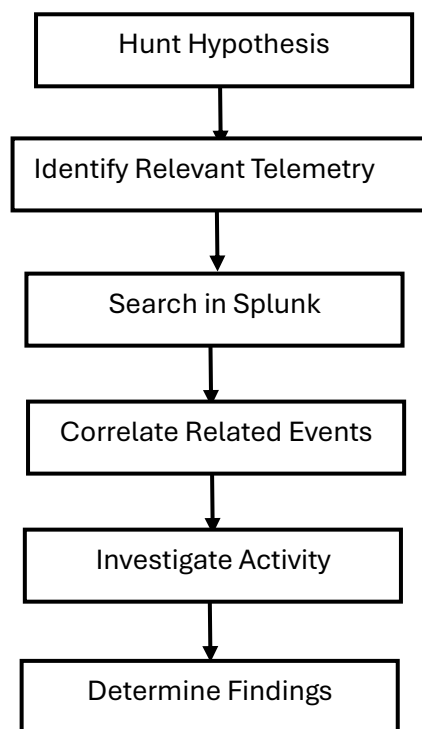
- Authentication and account activity
- Process creation and execution
- PowerShell activity
- DNS queries
- Network connections
- Endpoint security events
- Privileged-access changes

Splunk searches and dashboards provide a centralized view of this activity, enabling analysts to identify unusual events and correlate related activity rather than examining individual logs in isolation.

6.2 Threat Hunting

Threat hunting complements predefined detections by proactively searching available telemetry for suspicious behavior that may not have generated an alert. The process begins with a hypothesis about potentially malicious or abnormal activity and uses Splunk searches to investigate relevant endpoint data.

The hunting methodology follows:



Hunting areas within the project include PowerShell execution, process activity, DNS queries, network connections, account activity, and potential defense-evasion behavior.

6.3 Investigation Approach

During a hunt, the analyst examines the affected endpoint, user, timestamp, process information, command-line activity, and related events. Multiple events can be correlated to establish an activity timeline and determine whether the behavior is legitimate, suspicious, or requires further investigation.

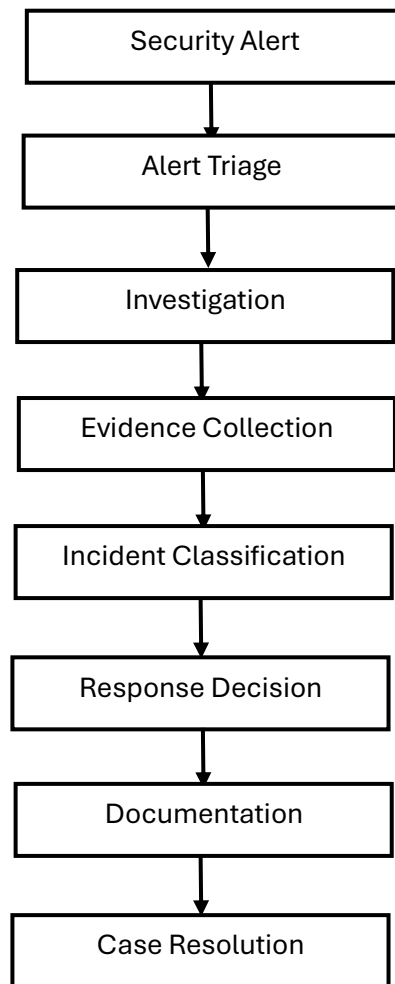
Detection-driven investigation begins with predefined detection logic, whereas threat hunting begins with an analyst hypothesis and searches available telemetry for supporting evidence. Together, monitoring and hunting provide both reactive and proactive security analysis capabilities within the CDOC.

07. Incident Response + TheHive

The CDOC follows a structured approach for moving from security detection to investigation and incident documentation. TheHive is used as the incident and case-management component, while Splunk provides the security telemetry and investigation evidence required by the analyst.

7.1 Incident Response Workflow

The incident-handling workflow follows:

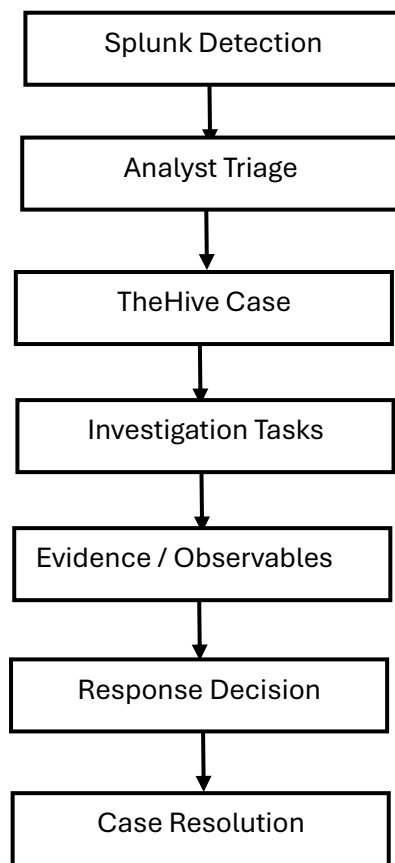


When suspicious activity is identified through Splunk monitoring or threat hunting, the analyst validates the activity and determines whether it represents legitimate behavior, a false positive, or a potential security incident. Relevant endpoint telemetry, including process activity, PowerShell execution, authentication events, DNS activity, network connections, and security events, is reviewed to understand the affected asset and potential impact.

7.2 TheHive Case Management

TheHive provides a structured location for documenting and managing security investigations. When an event requires formal incident handling, the investigation can be represented as a case containing incident information, investigation tasks, observables, evidence, analyst notes, and resolution details.

The workflow can be represented as:



This separates security telemetry analysis from incident case management. Splunk provides the underlying event data and investigation capabilities, while TheHive maintains structured case information and investigation records.

7.3 Investigation and Response

During investigation, the analyst correlates relevant telemetry to establish an activity timeline and assess the scope and severity of the observed activity. Where relevant, MISP can provide threat-intelligence context, while OpenVAS findings can provide vulnerability information about affected assets.

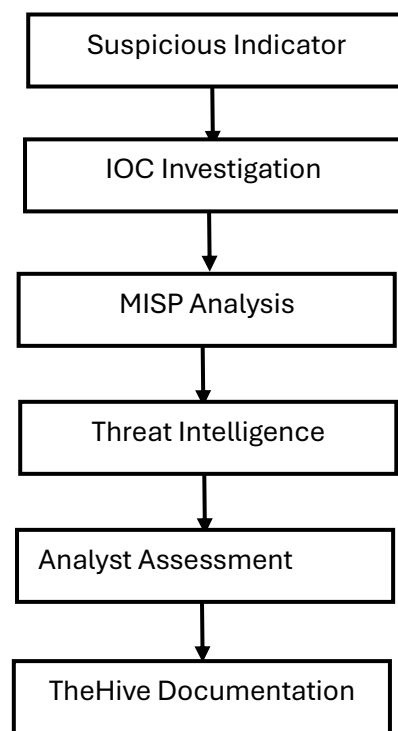
The case documentation captures the investigation outcome, supporting evidence, analyst findings, response decisions, and final case status. This demonstrates a structured SOC workflow in which security activity is triaged, investigated, documented, and managed through a defined case-management process.

08. MISP Threat Intelligence

MISP (Malware Information Sharing Platform) is included in the CDOC as the threat intelligence component of the SOC. Its purpose is to provide additional context for suspicious indicators identified during security monitoring and investigation. MISP complements Splunk and TheHive by supporting IOC investigation and threat intelligence analysis.

8.1 Threat Intelligence Workflow

The threat intelligence workflow follows:



Indicators identified during an investigation can include IP addresses, domains, URLs, and file hashes. MISP provides a structured environment for examining available information associated with these indicators and determining whether additional intelligence or context is available.

8.2 SOC Investigation

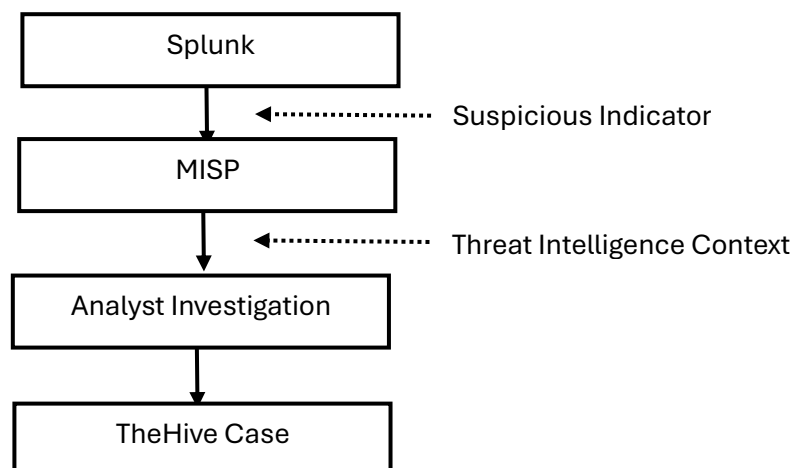
MISP is used as a supporting enrichment layer during security investigations. When an analyst encounters a suspicious indicator during a Splunk investigation, the indicator can be examined using available threat intelligence in MISP. This additional context can help determine whether the observed activity requires further investigation or escalation.

For example, when investigating a suspicious network connection, an analyst can identify the associated IP address from endpoint or network telemetry and examine the indicator in MISP. Relevant intelligence can then be considered alongside the original security telemetry when assessing the activity.

8.3 Integration with Incident Response

Threat intelligence can also support incident documentation through TheHive. Relevant indicators and intelligence identified during an investigation can be recorded as part of the incident case, connecting the original security event with the intelligence analysis and investigation outcome.

The workflow can be represented as:



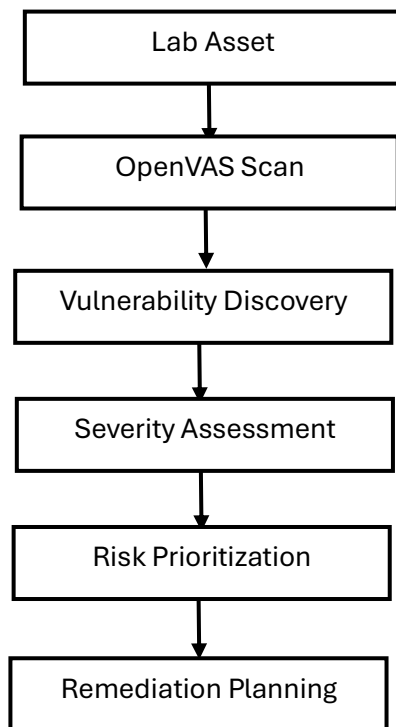
This workflow strengthens the CDOC's threat intelligence capability by incorporating IOC analysis and contextual intelligence into the broader SOC investigation process.

09. OpenVAS Vulnerability Management

OpenVAS is used in the CDOC as the vulnerability assessment component of the security operations workflow. Its purpose is to identify security weaknesses within the monitored lab environment and provide findings that support security analysis and remediation prioritization. OpenVAS complements the project's monitoring and detection capabilities by providing visibility into vulnerabilities affecting assessed assets.

9.1 Vulnerability Management Workflow

The vulnerability assessment process follows:



The process begins by identifying the asset to be assessed and configuring an OpenVAS scan. The scanner examines the target for known vulnerabilities and produces findings that can be reviewed by the security analyst.

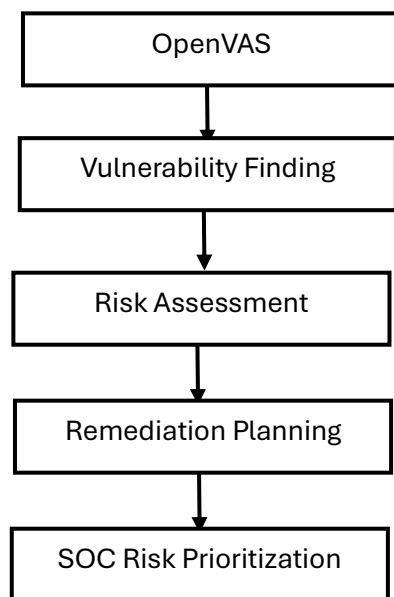
9.2 Vulnerability Analysis

The analyst reviews scan results to understand the affected asset, identified vulnerability, severity, and potential security impact. Findings can then be prioritized according to their severity and relevance to the security posture of the environment. Higher-severity vulnerabilities affecting important assets can receive greater remediation priority.

9.3 Integration with SOC Operations

OpenVAS complements the project's detection and investigation capabilities by providing a preventive security perspective. While Splunk provides visibility into security-relevant activity and endpoint events, OpenVAS helps identify weaknesses that could potentially be exploited.

The relationship can be represented as:



Vulnerability information can also provide additional context during security investigations. For example, if a security event affects an endpoint with a known vulnerability, the finding can help the analyst understand the potential risk associated with that asset.

9.4 Project Implementation

In this project, OpenVAS was used to perform vulnerability scanning against the lab environment and review the resulting security findings. The results contribute to the broader CDOC workflow by connecting asset assessment, vulnerability identification, severity analysis, and remediation planning.

This demonstrates that the CDOC extends beyond reactive monitoring by incorporating vulnerability assessment to identify weaknesses and support security prioritization within the monitored environment.

10. Results + Conclusion

10.1 Project Results

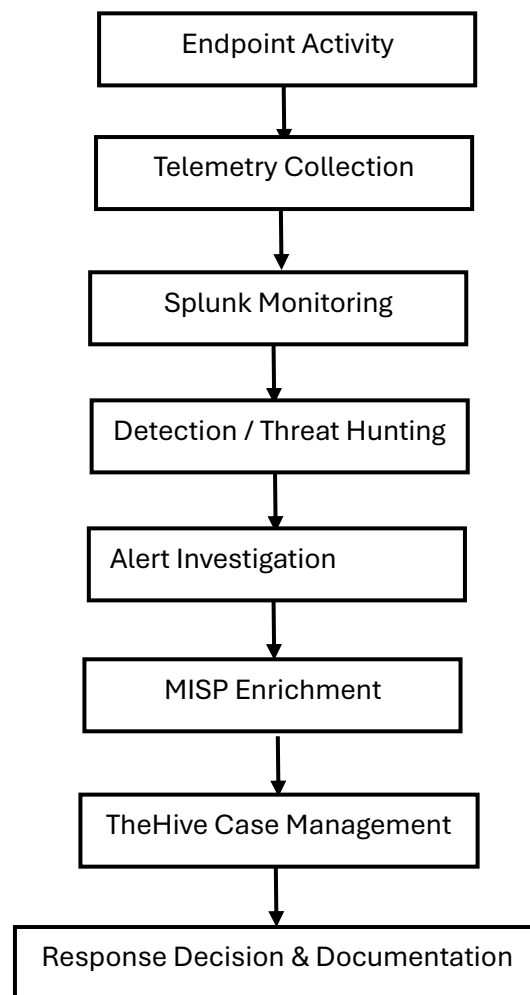
The CDOC established a virtualized SOC laboratory that brings together multiple security capabilities within a structured security operations workflow. Splunk Enterprise provides centralized monitoring and investigation using Windows telemetry collected through Sysmon, Windows Event Logs, and Microsoft Defender.

Capability	Implementation
SIEM Monitoring	Splunk Enterprise
Endpoint Telemetry	Sysmon + Windows Event Logs
Endpoint Security	Microsoft Defender
Network Analysis	Wireshark
Detection Engineering	Splunk-based detections
Threat Hunting	Splunk searches
Threat Intelligence	MISP
Case Management	TheHive
Vulnerability Assessment	OpenVAS

Implemented detections covered suspicious and encoded PowerShell activity, local account creation, administrator-group modifications, Defender-related tampering, process activity, DNS activity, and network connections.

10.2 SOC Workflow Demonstrated

The project demonstrates how the implemented capabilities support a structured SOC workflow:



Splunk provides centralized telemetry and investigation, MISP adds threat-intelligence context, TheHive supports case management, and OpenVAS provides vulnerability assessment.

10.3 Skills Demonstrated

The project provided practical exposure to centralized log analysis, alert triage, Splunk detection engineering, endpoint investigation, PowerShell and process analysis, threat hunting, IOC investigation, threat intelligence, incident documentation, vulnerability assessment, and security-tool usage within a virtualized environment.

10.4 Conclusion

The Enterprise Cyber Defense Operations Center (CDOC) demonstrates a practical, controlled SOC laboratory focused on security monitoring, detection, investigation, and supporting security operations capabilities rather than a single attack simulation.

By combining SIEM monitoring, endpoint telemetry, detection engineering, threat hunting, threat intelligence, case management, network analysis, and vulnerability assessment, the project demonstrates how multiple security technologies can contribute to a common SOC workflow. The implementation is intentionally presented as a laboratory environment rather than a production SOC and provides a foundation for future development toward automated response, expanded correlation, and enterprise-scale monitoring.